INCH-POUND
MIL-PRF-55365/12D
W/AMENDMENT 1
14 March 2008
SUPERSEDING
MIL-PRF-55365/12D
28 September 2007

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, POLARIZED ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY, STYLE CWR15

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-55365.

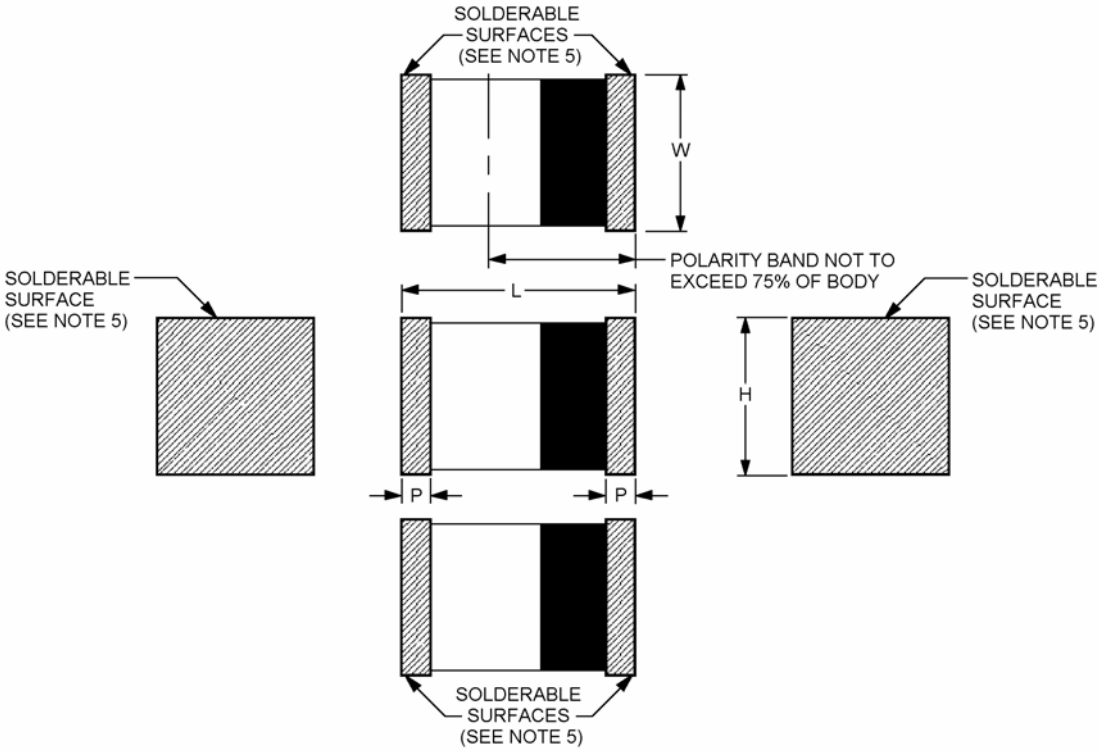


FIGURE 1. Dimensions and configuration.

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Case size	L + .010 (0.25) - .006 (0.15)	W + .008 (0.20) - .004 (0.10)	H + .008 (0.20) - .004 (0.10)	P + .014 (0.35) - .000 (0.0)
L	.063 (1.60)	.033 (0.85)	.033 (0.85)	.006 (0.15)
R	.079 (2.00)	.053 (1.35)	.053 (1.35)	.006 (0.15)
Case size	L ± .008 (0.20)	W ± .008 (0.20)	H ± .008 (0.20)	P + .014 (0.35) - .000 (0.0)
A	.126 (3.20)	.063 (1.60)	.063 (1.60)	.006 (0.15)
F	.158(4.01)	.116(2.95)	.050(1.27)	.010(0.15)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and given for general information only.
3. These capacitors are designed for mounting by dip soldering, reflow soldering, or other conventional means.
* 4. The anode (+) terminal shall be identified by a marking on all four sides or laser marking on one side of the case.
5. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.

FIGURE 1. Dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Termination finish: Termination finishes B, H, or K in accordance with MIL-PRF-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-PRF-55365).

Operating temperature range: -55°C to +125°C.

Product level designator: In accordance with MIL-PRF-55365 except the Weibull Failure Rate Level grading Failure Count during and after test shall be determined by DC leakage measurements of .0135 X CV or 0.1uA, whichever is greater.

DC leakage (DCL): See table I.

Capacitance: See table I.

Capacitance tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-PRF-55365 (see table I).

Resistance to soldering heat: In accordance with MIL-PRF-55365.

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Stability at low and high temperatures: In accordance with MIL-PRF-55365.

Step 1 (+25°C):	DCL	- See table I.
	Capacitance	- Shall not exceed tolerance of table I value.
	DF	- See table I.
Step 2 (-55°C):	Δ Capacitance	- Shall not exceed ±20 percent of initial measurement.
	DF	- See table I.
Step 3 (+25°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±10 percent of initial +25°C measurement.
	DF	- See table I.
Step 4 (+85°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±15 percent of initial +25°C measurement.
	DF	- See table I.
Step 5 (+125°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±20 percent of initial +25°C measurement.
	DF	- See table I.
Step 6 (+25°C):	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±10 percent of initial +25°C measurement.
	DF	- See table I.

Surge voltage: In accordance with MIL-PRF-55365.

DCL	- See table I.
Δ Capacitance	- Within ±5 percent of initial value.
DF	- See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C:	DCL	- See table I.
	Δ Capacitance	- Shall not exceed ±10 percent of initial measured value.
	DF	- See table I.
At 85°C:	DCL	- See table I.
At 125°C:	DCL	- See table I.

10,000 hours:

At 25°C:	DCL	- 200 percent of initial requirement.
	Δ Capacitance	- Shall not exceed ±10 percent of initial measured value.
	DF	- See table I.
At 85°C:	DCL	- 200 percent of initial requirement
At 125°C:	DCL	- 200 percent of initial requirement.

Solderability: In accordance with MIL-PRF-55365 except that following steam aging, test samples may have a 30 minute bake out at 150°C prior to solder dipping and all terminations shall exhibit a continuous solder coating free from defects for a minimum of 75 percent of the critical area of any individual termination.

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TABLE I. Style CWR15 characteristics.

Part or identifying number (PIN) 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (μF)	DC Leakage (max) (μA)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR15CK685--L-	4	6.8	.5	5	6	8	16	12	10	L
CWR15CK106--R-	4	10	.5	5	6	8	16	12	6	R
CWR15CK156--R-	4	15	.6	6	7	8	16	12	6	R
CWR15CK226--R-	4	22	.9	9	11	8	16	12	6	R
CWR15CK336--R-	4	33	1.3	13	16	10	20	15	6	R
CWR15CK686--A-	4	68	2.7	27	33	15	30	23	1	A
CWR15CK107--F-	4	100	4	40	48	20	40	33	1	F
CWR15DK335--L	6	3.3	.5	5	6	6	12	9	10	L
CWR15DK475--L-	6	4.7	.5	5	6	8	16	12	10	L
CWR15DK685--R-	6	6.8	.5	5	6	8	16	12	6	R
CWR15DK106--R-	6	10	.6	6	7	8	16	12	6	R
CWR15DK156--R-	6	15	.9	9	11	8	16	12	6	R
CWR15DK226--A-	6	22	1.4	14	17	10	20	15	6	A
CWR15DK336--A-	6	33	2.0	20	24	10	20	15	6	A
CWR15DK476--A-	6	47	2.8	28	34	15	30	23	4	A
CWR15DK686--F-	6	68	4	40	48	20	40	33	2	F
CWR15FK474--L-	10	0.47	.5	5	6	6	12	9	12	L
CWR15FK684--L-	10	0.68	.5	5	6	6	12	9	10	L
CWR15FK105--L-	10	1.0	.5	5	6	6	12	9	10	L
CWR15FK155--L-	10	1.5	.5	5	6	6	12	9	10	L
CWR15FK225--L-	10	2.2	.5	5	6	6	12	9	10	L
CWR15FK335--R-	10	3.3	.5	5	6	8	16	12	6	R
CWR15FK475--R-	10	4.7	.5	5	6	8	16	12	6	R
CWR15FK685--R-	10	6.8	.7	7	8.5	8	16	12	6	R
CWR15FK106--R-	10	10	1.0	10	12	8	16	12	6	R
CWR15FK156--A-	10	15	1.5	15	18	10	20	15	6	A
CWR15FK336--F-	10	33	3.3	33	40	15	30	25	6	F
CWR15HK105--L-	15	1.0	.5	5	6	6	12	9	18	L
CWR15HK475--R-	15	4.7	1.0	10	12	8	16	12	12	R
CWR15HK106--A-	15	10	1.5	15	18	10	20	15	12	A
CWR15HK156--F-	15	15	2.25	22.5	27	12	24	20	10	F

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TABLE I. Style CWR15 characteristics. Continued.

Part or identifying number (PIN) 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (μF)	DC Leakage (max) (μA)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR15JK474--L-	20	.47	.5	5	6	6	12	9	24	L
CWR15JK335--R-	20	3.3	.5	5	6	8	16	12	18	R
CWR15JK685--A-	20	6.8	1.4	14	17	10	20	15	18	A
CWR15JK106--F-	20	10	2.0	20	24	12	24	20	12	F
CWR15KK334--L-	25	.33	.5	5	6	6	12	9	30	L
CWR15KK155--R-	25	1.5	.5	5	6	6	12	9	24	R
CWR15KK335--A-	25	3.3	.8	8	9.6	6	12	9	18	A
CWR15KK475--F-	25	4.7	1.2	12	14.4	6	12	9	16	F
CWR15MK224--L-	35	.22	.5	5	6	6	12	9	35	L
CWR15MK474--R-	35	.47	.5	5	6	6	12	9	30	R
CWR15MK105--A-	35	1.0	.5	5	6	6	12	9	24	A
CWR15MK225--F-	35	2.2	.8	8	9.6	6	12	9	18	F
CWR15NK104--L-	50	.1	.5	5	6	6	12	9	40	L
CWR15NK334--R-	50	.33	.5	5	6	6	12	9	35	R
CWR15NK684--A-	50	.68	.5	5	6	6	12	9	30	A
CWR15NK155--F-	50	1.5	.75	7.5	9	6	12	9	24	F

1/ Complete PIN shall include additional symbols to indicate capacitance tolerance, product level designator, and surge current option letter. If surge current is not required, the last "-" shall be replaced with the letter Z.

NOTES:

Reference documents. In addition to MIL-PRF-55365, this specification sheet references the following documents.

MIL-STD-202

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

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Agent:
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